

VACCINATION DATA ANALYSIS AND VISUALIZATION

Professional Project Report

Public Health & Epidemiology | Python • SQL • Power BI

Executive Summary

This project analyzes the supplied vaccination coverage, disease incidence, reported cases, vaccine introduction and vaccine schedule datasets. It converts the source data into a reproducible analytical workflow and a Power BI-ready dashboard design.

The analysis is descriptive. Coverage–incidence relationships do not by themselves establish causal vaccine effectiveness.

Key 2023 indicators

Indicator	Result
Mean MCV1 coverage	85.21%
DTP1 → DTP3 drop-off	5.07 percentage points
MCV1 country observations	195
Correlation pairs	5

1. Methodology

Country-level records were used for analysis. Numeric measures were standardized to numeric types, missing values were not fabricated, and aggregate records were excluded from country-level calculations. The project is organized into cleaned data, analysis outputs, SQL design and Power BI visualization layers.

2. Exploratory Analysis

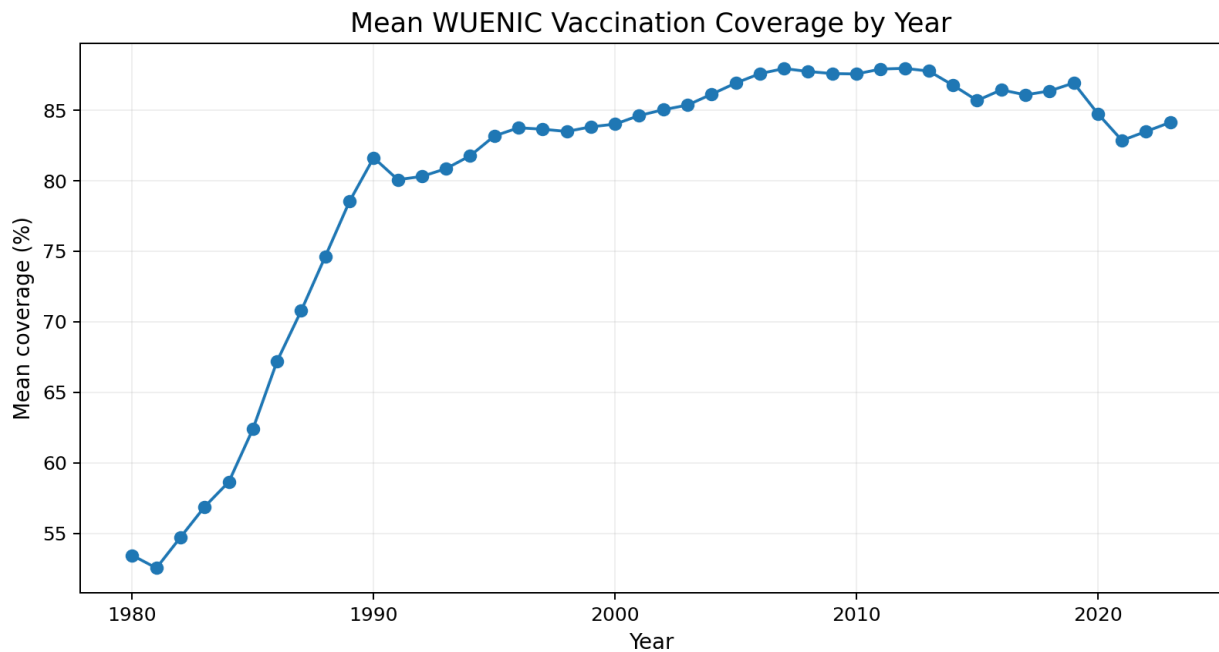


Figure 1. Mean WUENIC coverage trend.

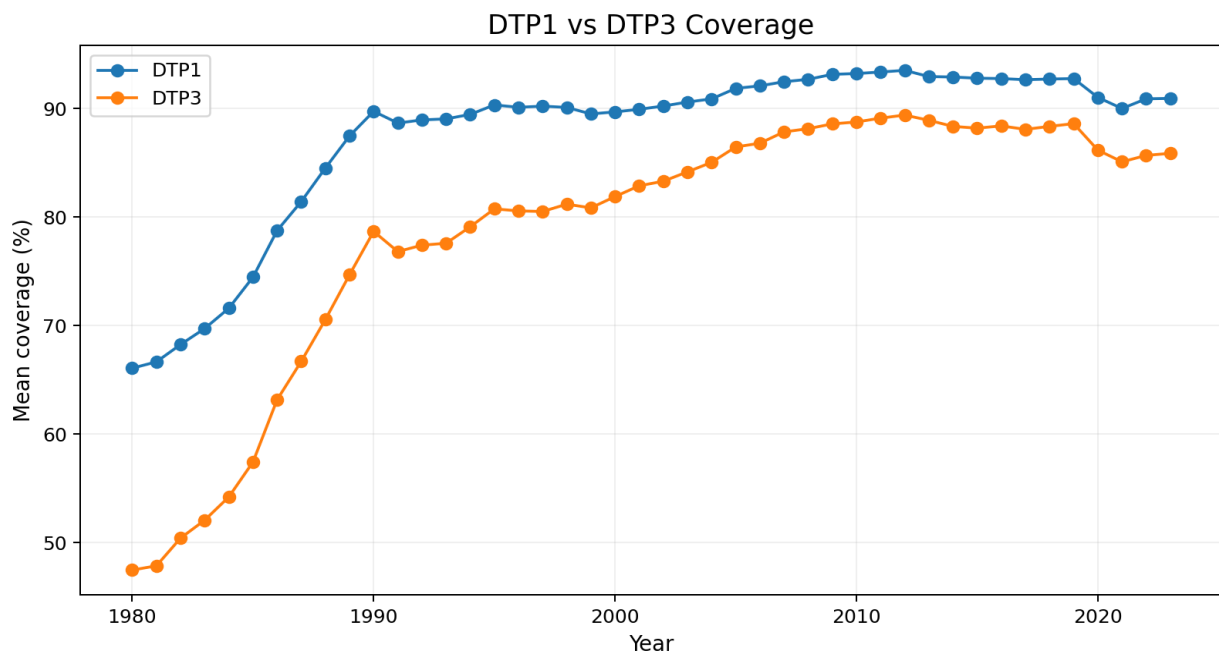


Figure 2. DTP1 vs DTP3 coverage.

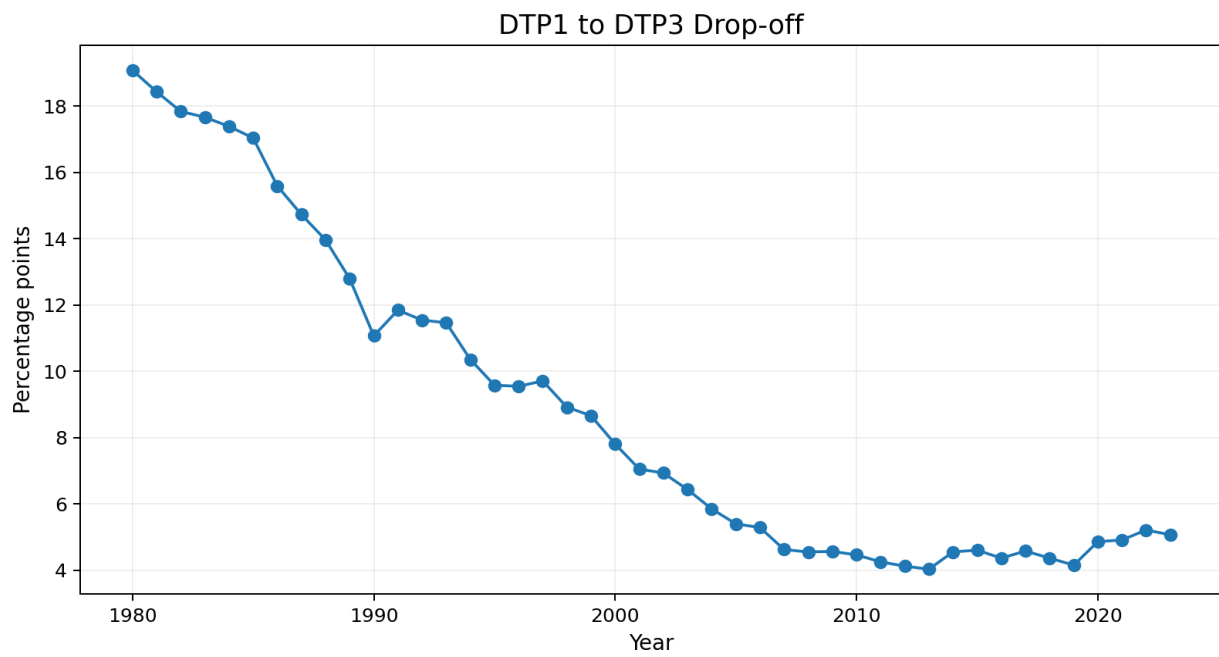


Figure 3. DTP1→DTP3 drop-off.

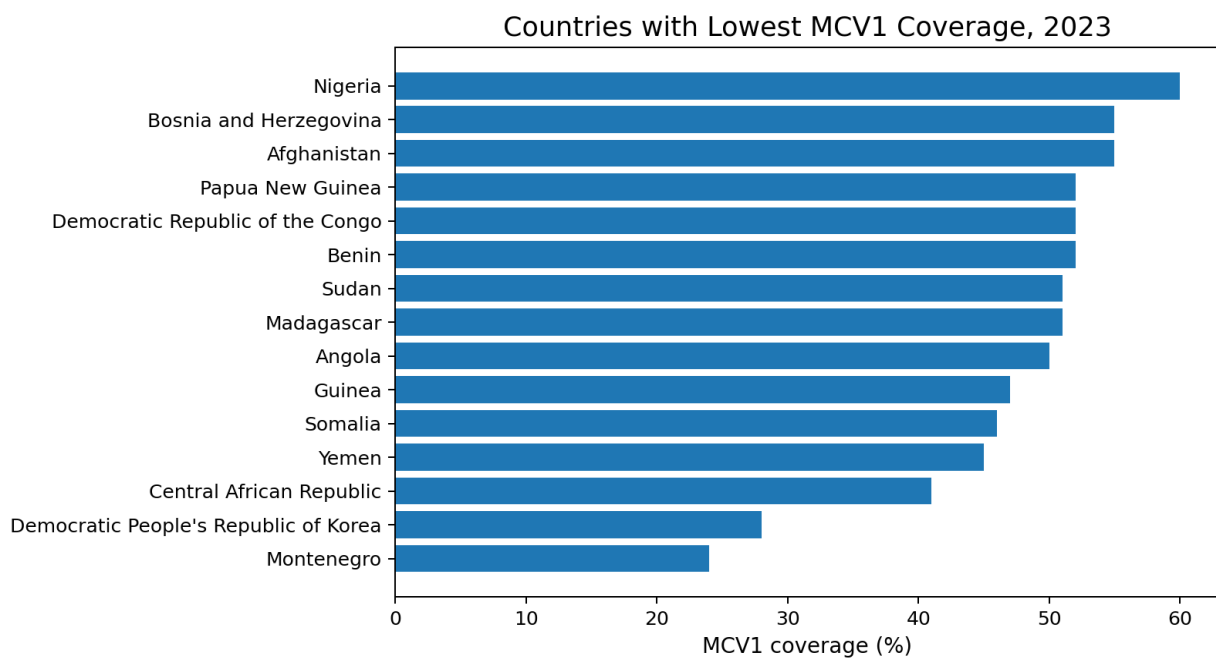


Figure 4. Lowest MCV1 coverage observations, 2023.

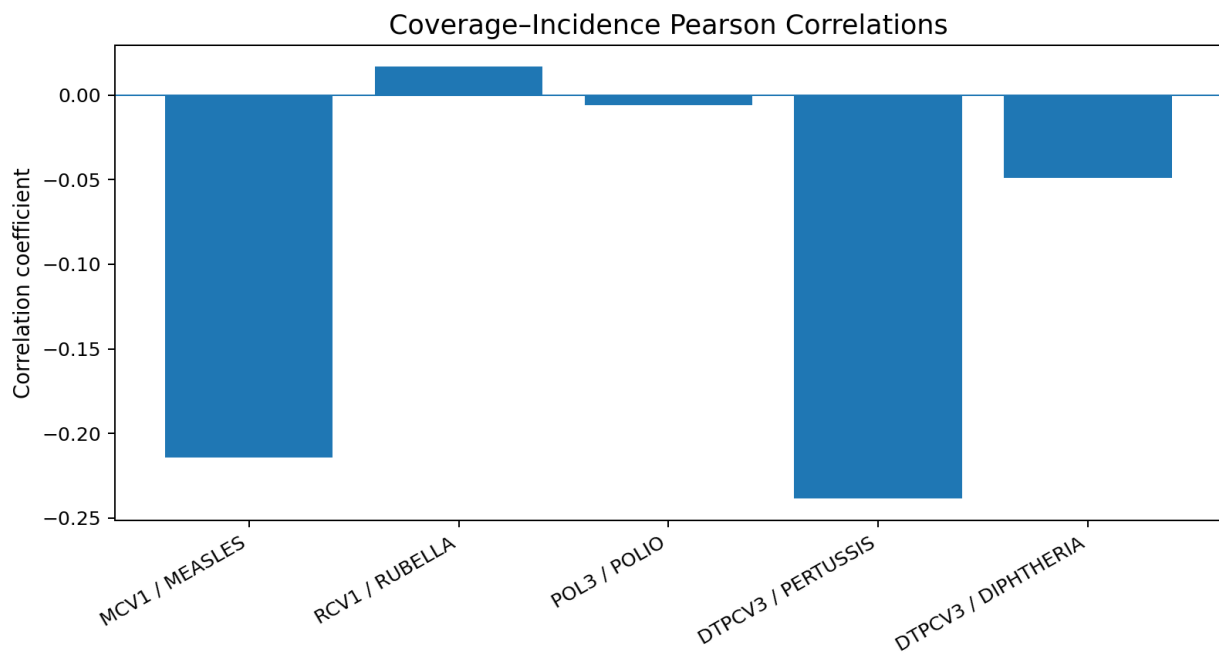


Figure 5. Descriptive coverage–incidence correlations.

High MCV1 Coverage and Measles Incidence, 2023

CODE	NAME	MCV1_COVERAGE_2023	INCIDENCE_RATE
AZE	Azerbaijan	96.0	1280.0
KGZ	Kyrgyzstan	96.0	996.1
KAZ	Kazakhstan	99.0	743.3
ZWE	Zimbabwe	90.0	338.5
IRQ	Iraq	97.0	273.6
ZMB	Zambia	90.0	272.8
ARM	Armenia	96.0	188.2
TON	Tonga	99.0	114.7
RUS	Russian Federation	97.0	90.0
SLE	Sierra Leone	90.0	76.2

Figure 6. High MCV1 coverage with measles incidence, 2023.

3. SQL Database Design

A normalized relational structure is supplied with country, year, disease and antigen dimensions and fact tables for coverage, reported cases, incidence, vaccine introduction and vaccine schedule. Quality checks and analytical queries are included.

4. Power BI Dashboard

The dashboard specification contains four professional report pages: Executive Overview; Vaccination Coverage; Disease Impact; and Vaccine Introduction & Schedule. KPI cards, trend charts, country comparisons, a scatter plot, introduction analysis and schedule analysis are specified.

5. Limitations

The supplied files do not contain gender, education, urban/rural classification, population density or monthly seasonality fields. Those questions therefore cannot be validly answered from the supplied datasets. Observational correlations should not be presented as causal effects.

6. Deliverables

This package includes cleaned data, analysis outputs, a professional Excel workbook, SQL package, Power BI build guide, figures and this report.

Conclusion

The resulting project is structured as a reproducible, dashboard-ready public-health analytics project suitable for academic submission and portfolio presentation.